Three-Year Dementia Risk Prediction from Multimodal NACC Data: Does Cardiovascular Information Add Value Beyond Cognition?

Team names and roll numbers

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Chapter 1

Introduction

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Chapter 2

Literature Review

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Chapter 3

Data

3.1 The National Alzheimer's Coordinating Center

The study uses the Freeze 74 release (June 2026) of the National Alzheimer's Coordinating Center (NACC) database. NACC pools longitudinal data from Alzheimer's Disease Research Centers across the United States. Volunteers are assessed roughly once a year with the Uniform Data Set (UDS), a fixed battery of questionnaires, examinations and neuropsychological tests that every centre administers in the same way [1]. The release used here contains 217,598 visits from 57,038 participants at 46 centres, collected between June 2005 and May 2026. Participants are referred through memory clinics or volunteer directly, so the cohort is a research sample and not a population sample: at the first visit the median age is 71 years, median education is 16 years and 76% of participants identify as White.

Access to the data is governed by the NACC data use agreement. All analysis was performed on a local machine. No participant-level value, identifier or image file name appears in this thesis, in the code repository or in any shared report; every reported number is an aggregate.

3.2 Files received

Table 3.1 lists the ten tables in the release. Only the main clinical table and the four summary PET tables contribute model inputs; the remaining files were audited but do not supply features.

Table 3.1: Files in the NACC release and their role in this study.

File	Rows	Participants	Role
Main clinical table (`ftldlbd`)	217,598	57,038	All clinical features and the outcome
SCAN amyloid PET summary	3,933	3,566	Centiloids and amyloid status
Mixed-protocol amyloid PET summary	2,235	2,073	Same, for older scans
SCAN tau PET	2,653	2,405	Meta-temporal and entorhinal SUVR
Mixed-protocol tau PET	671	616	Same, for older scans
SCAN and mixed-protocol amyloid PET, regional	6,168	—	Not used (tracer-specific SUVR)
SCAN FDG PET	885	639	Not used (about 600 linked participants)
SCAN PET quality-control list	8,759	4,585	Audit of the local image folders
CLARiTI study tracking	857	776	Not used (administrative)

In the clinical table one row is one visit; in the PET tables one row is one scan. The participant identifier is the only key shared by all tables. Because a scan is not tied to a visit number, each scan has to be matched to a visit by date (Section 4.3.4).

In addition, 599 raw PET acquisitions (318 amyloid, 232 tau and 49 FDG scans from 446 participants, about 100 GB of DICOM files) were available locally. They are used in Chapter 5.

3.3 The clinical table

The clinical table has 2,649 columns, but most of them are empty for most visits: modules for frontotemporal and Lewy body disease, autopsy findings, form version 4 items and free text. Only 787 columns contain data in at least 10% of visits. From these, 114 variables were mapped by hand to a meaning, a category, a role and their own missing-value codes, following the NACC Researcher's Data Dictionary. Seventy-six of them are used to build the model inputs.

3.3.1 Missing-value codes

NACC encodes non-responses with numeric codes whose meaning depends on the variable: -4 marks a question that was not part of the visit, codes such as 8, 88 or 888 mark "not applicable", and 9, 99 or 999 mark "unknown". These codes cannot be removed globally.

The value 88, for example, is a genuine age in 3,655 visits and a genuine diastolic pressure in 3,090 visits, but means "never smoked" in the years-smoked column (73,669 visits). Cleaning was therefore performed column by column from the variable map.

3.3.2 Changes in the forms

The UDS has been revised three times. The most important change for this study took place with version 3 in 2015, when the neuropsychological battery was replaced: the MMSE, Logical Memory, Digit Span and Boston Naming tests were replaced by the MoCA, Craft Story, Number Span and MINT tests. Only category fluency and the Trail Making tests are common to all versions. The same participant can therefore have different tests at different visits. Health history is also recorded on two forms: the participant-reported history (form A5) and, in version 3, the clinician-assessed conditions (form D2). Telephone visits contain no neuropsychological tests.

3.4 Feature categories

The variables used are grouped into five categories (Table 3.2). The complete list, with source columns and value codes, is given in Appendix A.

Table 3.2: Feature categories, source columns and number of model features.

Category	Main source columns	Features
Demographics	`NACCAGE`, `NACCSEX`, `EDUC`, `NACCNIHR`, `NACCHISP`, `NACCLIVS`	8
Cardiovascular and medical	Twelve conditions from forms A5 and D2, smoking, blood pressure, heart rate, BMI, four medication classes and the medication count	25
Cognitive tests	Twenty neuropsychological tests, the Geriatric Depression Scale and twelve NPI-Q behaviour items	9
Clinical stage	`CDRSUM`, `NACCUDSD`	3
Genetics	`NACCNE4S`, `NACCFAM`	3
PET (second-stage model only)	`CENTILOIDS`, amyloid status, meta-temporal and entorhinal tau SUVR	4

Four groups of variables were deliberately excluded. Autopsy findings are known only after death and would leak future information. The Functional Activities Questionnaire, the level of independence, subjective and informant-reported memory complaints and the use of Alzheimer's medication are so close to the diagnosis itself that including them would partly reproduce the clinician's decision; they were evaluated separately as a sensitivity analysis. Regional PET values were not used because they are tracer-specific and highly redundant. Form version 4 visits (3% of visits) were excluded because they use new variable names.

3.5 PET data

Amyloid PET values are available for about 5,100 participants who also appear in the clinical table, and tau PET values for 2,724. The amyloid tables provide the Centiloid value, which is comparable across tracers, and a positive or negative status assigned by the PET core. The tau tables provide standardised uptake value ratios (SUVR) in about 160 regions; the meta-temporal composite and the entorhinal cortex were retained as the standard summaries of early tau deposition. Because the two tau tracers (flortaucipir and MK-6240) give different SUVR scales, each tau value was standardised within tracer against amyloid-negative participants from the training set. Only scans that passed quality control were used.

Most PET scans are recent, so few scanned participants have three years of clinical follow-up after the scan. This limits what can be learned from PET in this cohort (Section 6.5).

Chapter 4

Methodology

4.1 Prediction task

The prediction is made at an *index visit*, using only information available at or before that visit. For a participant without dementia at the index visit, the task is to estimate the probability of a clinical diagnosis of dementia within three years:

$$\hat{p} = P\left(\text{dementia diagnosed in } (t_0, t_0 + 3 \text{ years}] \mid \mathbf{x}_{t_0}\right), \tag{4.1}$$

where t_0 is the date of the index visit and $\mathbf{x}_{t_0}$ the feature vector at that visit. The main analysis uses each participant's first visit as the index visit.

4.1.1 Choice of horizon and outcome

Clinical diagnosis (`NACCUDSD`) and the Clinical Dementia Rating are recorded at every visit in every form version, which makes them the most consistent outcomes available. A dementia diagnosis was chosen as the main outcome because it is clinically meaningful and easy to interpret. The three-year horizon balances the number of events against loss to follow-up: Kaplan–Meier estimates show that about 35% of participants with MCI progress to dementia within three years, and that 11% of cognitively normal participants progress to MCI or dementia, while the number of participants with a known status falls steadily with longer horizons and most PET scans are too recent for a five-year horizon.

4.1.2 Censoring

A participant last seen less than three years after the index visit without a dementia diagnosis has an unknown outcome. Such participants were excluded from the binary task rather than counted as negative, which would bias the estimated risk downwards. Because this exclusion could itself introduce bias, a Cox proportional hazards model that uses all eligible participants, including the censored ones, was fitted as a robustness check (Section 6.8).

4.2 Cohort

Table 4.1 shows how the cohort was formed. Visits had to be in person and on form versions 1 to 3, because telephone and version 4 visits lack the neuropsychological tests. Participants with dementia at the index visit, or at any earlier visit, are not eligible.

Table 4.1: Cohort flow.

Step	Participants
All participants	57,038
First visit in person, form version 1 to 3	55,461
and not demented at that visit	38,107
and outcome known at three years	19,470
of whom diagnosed with dementia within three years	2,818 (14.5%)

The "not demented" cohort contains three clinically different groups: 12,377 cognitively normal participants (198 converters, 1.6%), 5,993 participants with MCI (2,485 converters, 41.5%) and 1,100 participants classed as impaired but not MCI (135 converters, 12.3%). Results are reported for the whole cohort and for the normal and MCI groups separately.

4.3 Feature construction

4.3.1 Harmonised cognitive scores

To place tests from different form versions on one scale, each raw score s was converted into a demographically adjusted z-score. For each test, a linear model of the score on age, sex and education was fitted among cognitively normal participants at their first visit, using the training set only:

$$\hat{s} = \beta_0 + \beta_1\,\mathrm{age} + \beta_2\,\mathrm{female} + \beta_3\,\mathrm{education}, \qquad z = \frac{s - \hat{s}}{\sigma_{\mathrm{res}}}, \tag{4.2}$$

where σ_{res} is the residual standard deviation of the fitted model. The sign is reversed for the two timed Trail Making tests, so that a negative z always means worse than expected. Values were clipped to $[-5, 5]$. The z-scores of the tests in one cognitive domain were then averaged to give six domain scores: global screening (MoCA, MMSE), memory, attention, speed and executive function, language, and visuospatial function. A participant receives a domain score from whichever tests their visit contained.

As an example, for the MoCA the fitted model is $\hat{s} = 23.0 - 0.059\,\mathrm{age} + 0.33\,\mathrm{female} + 0.42\,\mathrm{education}$ with $\sigma_{\mathrm{res}} = 2.56$, estimated from 6,264 normal participants. A 72-year-old woman with 16 years of education has an expected score of 25.8, so a score of 21 corresponds to $z = -1.9$.

Three further cognitive features were derived: the number of tests the participant could not complete for cognitive or behavioural reasons (NACC code 96 or 996), which is informative about severity and is therefore kept rather than treated as missing; the Geriatric Depression Scale total; and the number of NPI-Q behavioural symptoms present.

4.3.2 Cardiovascular history

Each condition was coded as "ever present" from the participant-reported history (form A5, where both recent and remote mentions count) and, where that form was blank, from the clinician-assessed form D2. Once present, a condition remains present at all later visits. Measured blood pressure, heart rate and BMI use the most recent measurement up to the visit. Pulse pressure and a count of four vascular risk factors (hypertension, hypercholesterolaemia, diabetes and smoking) were derived.

4.3.3 Genetics

The number of APOE $\varepsilon 4$ alleles is used directly. About one participant in five was not genotyped; genotyping depends on centre and consent and is therefore not random, so these participants receive a separate "not genotyped" indicator instead of an imputed value.

4.3.4 Preventing the use of future information

No feature at a visit may depend on a later visit. Fixed characteristics such as education are carried forward from the visit where they were first recorded, never backwards. A PET scan is used at a visit only if it was acquired on or up to 365 days before that visit. Both rules were introduced after the verification step described in Chapter 8 found violations in an earlier version of the pipeline.

4.4 Data splits

All splits were made at participant level, so that all visits of a person fall in the same partition. Twenty percent of the centres (9 of 46) were held out entirely to measure performance at sites the model has never seen. The remaining participants were divided into training (70%), validation (15%) and test (15%) sets, stratified by diagnosis at the first visit. The random seed was fixed (2026). Table 4.2 gives the resulting sizes for the first-visit cohort.

Table 4.2: Partition sizes for the main cohort.

Partition	Participants	Converters
Training	10,879	1,599
Validation	2,241	342
Test	2,286	332
Held-out centres	4,064	545

Every quantity estimated from data (normative models for z-scores, tau standardisation, imputation medians, scaling) was fitted on the training set and applied unchanged to the other sets. The validation set was used for early stopping and model selection; the test set and the held-out centres were used only for evaluation.

4.5 Models

Five classifiers of increasing complexity were compared on the same 48 features (Table 4.3). Logistic regression is the standard baseline in clinical prediction. Random forests [2] and gradient-boosted trees (XGBoost [3]) are strong learners for tabular data that handle non-linear effects and interactions. A multilayer perceptron and the FT-Transformer [4] represent deep learning; the latter tokenises each feature and applies self-attention, and is among the strongest deep architectures reported for tabular data. Except for logistic regression, every model was trained with three random seeds and the predictions were averaged.

Table 4.3: Models and settings.

Model	Settings
Logistic regression	Median imputation with missing-value indicators, standardisation; regularisation strength chosen from five values on the validation set
Random forest	500 trees, minimum 10 samples per leaf, $\sqrt{p}$ features per split
XGBoost	Up to 3,000 trees, learning rate 0.03, depth chosen from {3, 4, 6} on validation AUROC, subsample and column sample 0.8, early stopping after 100 rounds without improvement; missing values handled natively
Multilayer perceptron	Two hidden layers (128 and 64 units), dropout 0.2, AdamW, early stopping on validation AUROC
FT-Transformer	Feature tokens of dimension 64, three Transformer layers with eight heads, a learned token for each missing feature, early stopping on validation AUROC

XGBoost was used as the main model for all further experiments because it performed as well as the others, handles missing values without imputation and allows exact Shapley value explanations.

4.6 Contribution of each feature category

The contribution of a category was measured by ablation: the model was trained with all categories and again without the category of interest, and the paired difference in AUROC on the same participants was computed with bootstrap confidence intervals. Each category was also evaluated alone and in cumulative order.

4.7 Longitudinal models

To test whether earlier visits add information, the index visit was moved to a later "landmark" visit preceded by at least two earlier visits. Three models were compared on the same landmarks: XGBoost on the landmark visit only; XGBoost with slope and change features of the cognitive scores and CDR-SB over the earlier visits; and a gated recurrent unit (GRU) [5] with 48 hidden units that reads up to six visits, with the time of each visit as an input because visit intervals are irregular.

4.8 PET late fusion

Because few scanned participants have three years of follow-up, a large model cannot be fitted on PET participants alone. A late-fusion design was used: the clinical model, trained on all participants, produces a risk for each scanned participant; a small, strongly regularised logistic regression then combines the logit of that risk with the PET measures and indicators of which measures are present. Two rules for matching scans to visits were compared: a strict rule (scan on or before the index visit) and a nearest-visit rule (larger, but the scan may follow the visit by a few weeks).

4.9 Modality dropout

In practice not every patient has every assessment. A model trained on complete records may behave poorly when an entire category is missing. To make one model usable with any combination of available information, the training data were augmented with four additional copies in which each optional category (genetics, cardiovascular, cognitive tests, clinical stage) was removed independently with probability 0.35. This model was compared with a standard model and with specialist models retrained for each pattern of missing categories.

4.10 Explainability

Predictions were explained with Shapley additive explanations (SHAP) [6], computed exactly for the tree ensemble with TreeSHAP [7]. For each test participant, SHAP decomposes the predicted log-odds into a sum of feature contributions. Global importance is the mean absolute contribution of a feature over the test set; category importance is the sum over its features. Stability was assessed by refitting the model on eight bootstrap samples of the training set.

4.11 Evaluation

Discrimination was measured by the area under the ROC curve (AUROC) and the area under the precision–recall curve (AUPRC). Calibration was assessed by the Brier score, calibration slope and intercept, and by comparing predicted and observed risk in deciles. Clinical usefulness was assessed by decision curve analysis [8] and by sensitivity, specificity and predictive values at fixed risk thresholds. Uncertainty was quantified by 1,000 bootstrap resamples of the evaluation set; differences between models were computed on the same resamples (paired bootstrap). Performance was examined within subgroups defined by sex, age, race, education, APOE status and baseline diagnosis.

Chapter 5

PET Image Pipeline

This chapter describes a separate experiment on the 599 raw PET acquisitions. Its purpose is to establish whether the images can be processed into reliable quantitative measurements and used by an image model. It does not feed the dementia model, for the reason given in Section 5.6.

5.1 From DICOM series to static volumes

Each acquisition arrives as a dynamic DICOM series of between 188 and 4,134 files: the same volume repeated over several time frames, from about 20 scanner models. For every scan the stored values were converted to activity concentration, slices were grouped into frames and ordered spatially, and the frames whose mid-point fell inside the tracer's standard static acquisition window were averaged, each weighted by its duration. The result is one static three-dimensional volume per scan. The raw files were only read; all outputs were written to a separate directory.

5.2 Spatial registration

Scans differ in matrix size, voxel spacing and head position, so the same voxel does not correspond to the same anatomical location across scans. No structural MRI was available, so a PET-only registration in two passes was used. In the first pass each scan was aligned to the MNI152 T1 template with a rigid transform (rotation and translation), using mutual information as the similarity measure because PET and T1 images have different contrast. The aligned scans of each tracer class were averaged to form a study-specific PET template, and in the second pass each scan was aligned to that template with a full affine transform (12 parameters). Aligning PET to a PET template is more robust than aligning it to an MRI template.

5.3 Amyloid SUVR from the images

For each registered amyloid scan, the standardised uptake value ratio was computed as the mean uptake in the cortical target region divided by the mean uptake in the

whole cerebellum, using the standard masks of the Centiloid project [9]. NACC's PET core had computed an SUVR for the same scans with a different, MRI-based pipeline in native space. Agreement between the two therefore provides an independent test of the conversion and registration steps.

Table 5.1: Agreement between image-derived and NACC amyloid SUVR (318 scans).

Measure	Value
Pearson correlation	0.977
Spearman correlation	0.885
Regression slope (NACC on ours)	1.10
Mean difference (ours minus NACC)	−0.043
95% limits of agreement	−0.18 to +0.09

Agreement was high for every tracer (Pearson r between 0.96 and 0.99 for the four tracers with known identity) and every scanner manufacturer. The lower Spearman correlation reflects the many amyloid-negative scans with similar low values, where small differences change the rank order.

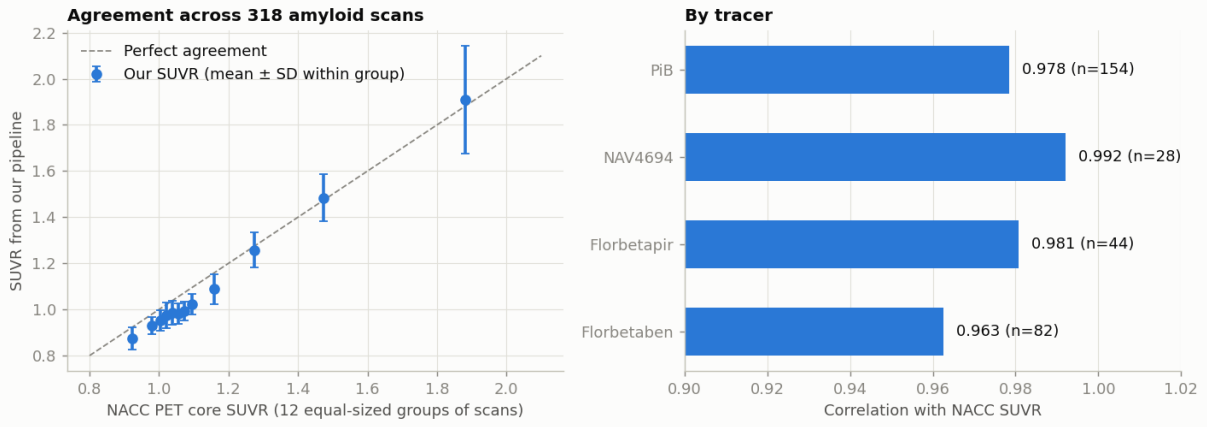


Figure 5.1: Image-derived amyloid SUVR against the value computed by the NACC PET core for the same scans.

5.4 Amyloid status classification

Three classifiers of amyloid status were compared on the 292 amyloid scans with a status label from the PET core (84 positive), using five-fold stratified cross-validation so that every scan was predicted by a model that had not seen it: (i) the cortical SUVR alone; (ii) a logistic regression on mean uptake in an $8 \times 8 \times 8$-voxel grid of blocks; and (iii) a small 3D convolutional neural network. The network has four stages of two $3 \times 3 \times 3$ convolutions with instance normalisation and ReLU, each stage halving the resolution, followed by global average pooling, dropout of 0.3 and a linear output, with 494,917 weights in total. Volumes were cropped to the head and resampled to $80 \times 96 \times 80$ voxels. Training used AdamW (learning rate 10^{-3}, weight decay 10^{-2}) with a one-cycle schedule, 40 epochs, batches of 8, a class-weighted loss, and augmentation by left–right flips, shifts of up to three voxels and small intensity changes.

Table 5.2: Amyloid status classification (292 scans, 84 positive, cross-validated).

Method	AUROC
Cortical SUVR (one number)	0.982
Regional logistic model	0.971
3D CNN	0.960

The single SUVR value performed best. This is expected: the reference label is itself derived by the PET core from a cortical uptake measure, so a scalar SUVR is close to the definition of the label, and a network trained on fewer than 300 scans has little opportunity to improve on it. The experiment shows that the image pipeline supports a working deep-learning model, not that a CNN is preferable for this task.

5.5 Grad-CAM

Gradient-weighted class activation maps [10] were computed for the network and averaged over 72 scans. The maps were diffuse: about 96% of the volume exceeded a quarter of the maximum activation, and only 1.7% of the most strongly activated voxels lay inside the cortical target region, against 3.8% expected by chance. The maps therefore do not show that the network attends to the cortical regions where amyloid accumulates, and they are not used as evidence of what the model has learned.

5.6 Why images are not used to predict dementia

Of the 446 participants with an image, about 86% were cognitively normal at the time of the scan, follow-up after the scan is about two years, and only five later received a dementia diagnosis. No model can learn a reliable pattern from five events, so dementia prediction from images was not attempted. The quantitative PET information that is available for thousands of participants is used instead, through the late-fusion model described in Chapter 4.

Two further limitations apply. Registration without an MRI is less precise than the MRI-based pipelines used by PET cores. Tau images were converted and registered but not quantified, because a standard tau target mask comparable to the Centiloid masks was not available.

Chapter 6

Results

All results in this chapter are computed on participants who were not used for training: the internal test set (2,286 participants, 332 converters) and the nine held-out centres (4,064 participants, 545 converters). Intervals are 95% bootstrap intervals. Every number can be traced to a result file in the project repository.

6.1 Comparison of models

Table 6.1: Discrimination and calibration of the five models.

	Test set			Held-out centres	
Model	AUROC (95% CI)	AUPRC	Brier	AUROC	Brier
Logistic regression	0.936 (0.924–0.949)	0.721	0.066	0.945	0.060
Random forest	0.932 (0.919–0.944)	0.704	0.070	0.940	0.064
XGBoost	0.938 (0.925–0.948)	0.713	0.066	0.946	0.059
Multilayer perceptron	0.938 (0.926–0.949)	0.721	0.066	0.946	0.059
FT-Transformer	0.939 (0.927–0.949)	0.720	0.066	0.945	0.061

The five models perform almost identically, and their confidence intervals overlap almost completely (Table 6.1). The deep-learning models did not outperform logistic regression or gradient boosting on these data. For reference, always predicting the training event rate gives a Brier score of 0.124, so all models are substantially better calibrated than a constant prediction.

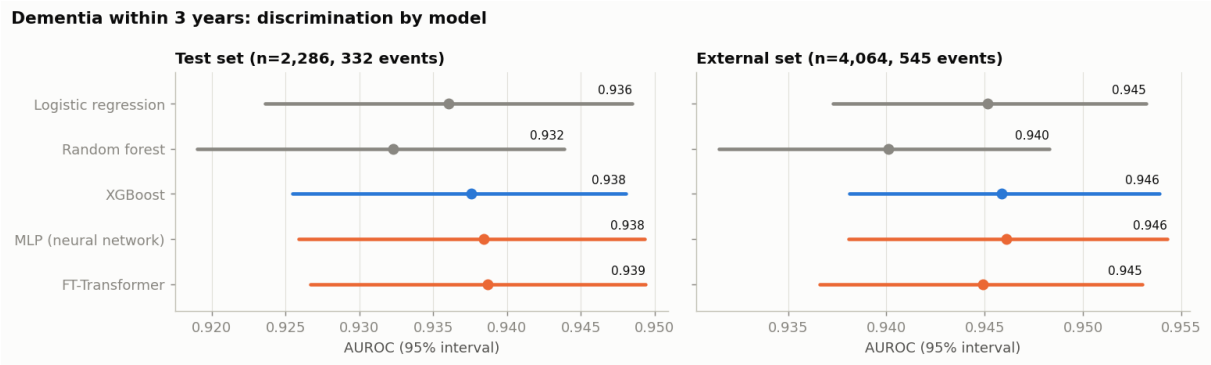


Figure 6.1: AUROC and AUPRC of the five models on the test set and the held-out centres.

6.2 Performance by diagnosis at the index visit

The cohort combines cognitively normal participants, who rarely convert (1.6%), with participants with MCI, who often do (41.5%). Diagnosis alone therefore separates converters from non-converters well: a model given only the diagnosis reaches an AUROC of 0.84. Table 6.2 reports XGBoost separately within each group.

Table 6.2: XGBoost by diagnosis at the index visit (participants / converters).

	Test set		Held-out centres	
Group	n / events	AUROC (95% CI)	n / events	AUROC (95% CI)
All not demented	2,286 / 332	0.937 (0.925–0.948)	4,064 / 545	0.946 (0.938–0.954)
Cognitively normal	1,411 / 24	0.884 (0.819–0.936)	2,624 / 37	0.876 (0.823–0.918)
MCI	732 / 289	0.835 (0.807–0.862)	1,291 / 496	0.846 (0.824–0.867)

Within MCI the AUROC is about 0.84, which is the more informative estimate of how well the model ranks risk among people who are already impaired. Within the cognitively normal group the AUROC is about 0.88, but it rests on only 24 and 37 events and the intervals are wide; with an event rate of 1.6% the average precision is only 0.15 to 0.19, so most participants flagged as high risk would not convert. Models trained only on normal participants (AUROC 0.876 and 0.881) or only on MCI participants (0.829 and 0.838) gave the same picture, so the group results are not an artefact of mixing the groups during training.

6.3 What drives the prediction

6.3.1 Feature categories

Table 6.3: Contribution of each feature category (XGBoost, test set unless stated).

Category	Alone	Gain when added to all others (test)	Gain (held-out centres)
Cognitive tests	0.896	+0.022 (+0.015 to +0.029)	+0.019 (+0.013 to +0.02
Clinical stage (CDR-SB, diagnosis)	0.897	+0.015 (+0.007 to +0.023)	+0.021 (+0.015 to +0.02
Demographics	0.671	+0.010 (+0.006 to +0.015)	+0.009 (+0.006 to +0.01
Genetics (APOE)	0.645	+0.003 (+0.002 to +0.004)	+0.001 (−0.000 to +0.00
Cardiovascular and medical	0.626	+0.001 (−0.000 to +0.003)	+0.001 (+0.000 to +0.00

Cognitive tests and clinical stage each reach an AUROC near 0.90 on their own and make the largest unique contributions (Table 6.3, Figure 6.2). Because these two categories overlap, removing either one costs little; removing both together, so that the model has no information about current cognition, lowers the AUROC from 0.937 to 0.746 on the test set (0.744 on the held-out centres). Within the normal and MCI groups the corresponding values are 0.783 and 0.638. The model therefore relies mainly on how impaired a person already is at the index visit. Because all these variables are recorded at the index visit and the outcome occurs later, this is not leakage, but it means the model is best described as estimating short-term progression given the current clinical stage.

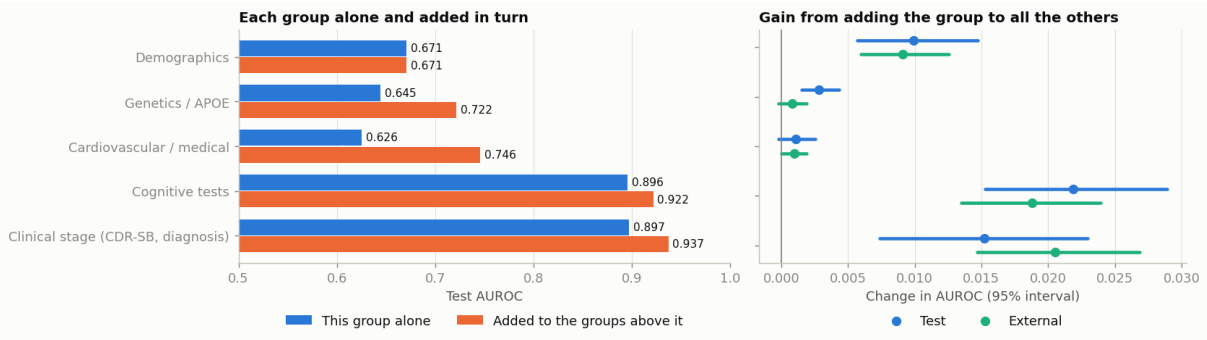


Figure 6.2: Left: AUROC of each category alone and added cumulatively. Right: gain in AUROC from adding each category to all the others.

6.3.2 Cardiovascular information (research question 1)

On its own, the cardiovascular category carries some signal (AUROC 0.626). Added to demographics and APOE only, before any cognitive information, it improved the AUROC by 0.024 on the test set (0.011 to 0.039), but by only 0.004 at the held-out centres (−0.006 to 0.015). Once cognitive information is present its contribution falls to about 0.001, with an interval that includes zero on the test set. Within the cognitively normal group the gain was 0.008 on the test set and 0.003 at the held-out centres, with intervals spanning zero. In this cohort and prediction setting, therefore, cardiovascular and medical history added little incremental predictive value beyond current cognitive status. This finding does not imply that vascular health is unimportant for the brain: the participants enter

at a median age of 71, mid-life exposure is not observed, and vascular damage that has already occurred is likely to be reflected in the cognitive scores.

6.3.3 Shapley explanations

Table 6.4: Share of mean absolute SHAP value by category (test set, 2,286 participants).

Category	Share
Clinical stage (CDR-SB, diagnosis)	40.8%
Cognitive tests	35.0%
Demographics	15.5%
Cardiovascular and medical	5.1%
Genetics	3.5%

The most important individual features were CDR-SB, MCI status, the memory score, age and the executive score. The ranking was stable across eight bootstrap refits (Spearman correlation with the main model 0.89 to 0.97; 8 to 10 of the top ten features shared). Some associations are not causal: lower BMI was associated with higher predicted risk, which is consistent with weight loss in the years before a dementia diagnosis rather than a protective effect of higher weight.

6.4 Longitudinal information

On landmark visits preceded by at least two earlier visits (test: 1,205 participants, 166 events; held-out centres: 2,254 participants, 326 events), adding the visit history produced only small changes (Table 6.5).

Table 6.5: Effect of earlier visits (paired gain over the landmark-only model).

Model	Test AUROC (gain)	Held-out AUROC (gain)
XGBoost, landmark visit only	0.919	0.953
XGBoost with trajectory features	0.922 (+0.003; −0.006 to +0.011)	0.956 (+0.003; −0.001 to +0.007)
GRU over the visit sequence	0.920 (+0.000; −0.011 to +0.010)	0.958 (+0.005; +0.001 to +0.008)

When current scores are known, the path that led to them adds little for a three-year prediction. This does not show that longitudinal modelling is of no value; a longer horizon, an earlier disease stage or more frequent measurement could give a different result.

6.5 PET measures (research question 2)

On its own, amyloid status is strongly associated with progression: among participants with MCI in the held-out sets, about 34% of amyloid-positive participants received a dementia diagnosis within two years, compared with about 7% of amyloid-negative participants (36% and 9% under the strict scan-matching rule). Added to the clinical model,

however, the PET measures did not produce a clear improvement under either matching rule. For example, with strict matching and a three-year horizon, the AUROC at the held-out centres was 0.939 without and 0.946 with amyloid (191 participants, 27 events; paired gain −0.007 to +0.024). The final late-fusion model scored 0.964 against 0.944 for the clinical model on 243 held-out scanned participants with 33 events, a gain of 0.020 whose interval (−0.009 to +0.066) includes zero. A modest benefit cannot be excluded, but it could not be demonstrated with the follow-up currently available.

6.6 Validation

6.6.1 Subgroups

Table 6.6: XGBoost AUROC by subgroup.

Subgroup	Test (n / events)	Held-out centres (n / events)
Female	0.950 (1,338 / 162)	0.947 (2,507 / 272)
Male	0.919 (948 / 170)	0.941 (1,557 / 273)
Age under 65	0.976 (467 / 46)	0.958 (1,067 / 56)
Age 65–74	0.952 (963 / 99)	0.956 (1,463 / 191)
Age 75–84	0.904 (732 / 157)	0.937 (1,044 / 191)
Age 85 and over	0.852 (124 / 30)	0.888 (490 / 107)
White	0.933 (1,813 / 273)	0.943 (3,347 / 499)
Black	0.948 (304 / 31)	0.951 (471 / 26)
Education 12 years or less	0.926 (400 / 92)	0.920 (744 / 137)
Education over 16 years	0.951 (937 / 112)	0.960 (1,551 / 188)
APOE $\varepsilon 4$ carrier	0.915 (762 / 150)	0.933 (1,402 / 277)

Discrimination was similar across sex, race and education. It was lower in the oldest participants and in APOE $\varepsilon 4$ carriers, both groups with a higher baseline risk.

6.6.2 Calibration and clinical usefulness

On the test set the mean predicted risk (14.5%) equalled the observed rate (14.5%), with a calibration slope of 1.01 and intercept of 0.009. At the held-out centres the model over-predicted slightly (15.0% predicted, 13.4% observed; slope 1.10). Using a risk threshold of 20%, the model flagged 22% of test participants with a sensitivity of 83%, specificity of 88%, positive predictive value of 55% and negative predictive value of 97% (held-out centres: 90%, 87%, 52% and 98%). Decision curve analysis showed a higher net benefit than treating everyone or no one across all thresholds examined, from 2% to 60%, on both evaluation sets.

6.7 Missing information

Table 6.7: Brier score when categories are missing (test set; lower is better).

Available information	Standard model	Modality dropout	Specialist model
Everything	0.066	0.067	0.066
No cognitive tests	0.082	0.077	0.080
No clinical stage	0.185	0.076	0.074
No cognitive information at all	0.441	0.115	0.111
Demographics only	0.444	0.120	0.118

A standard model given missing inputs at prediction time produced badly calibrated risks when cognitive information was absent, worse than predicting the base rate (Brier 0.124). The model trained with modality dropout stayed close to a specialist model retrained for each situation (Table 6.7), with almost no loss when all information is present. This model is used in the clinician interface.

6.8 Sensitivity analyses

Censored participants. A Cox model fitted on all eligible participants, including those followed for less than three years, reached a concordance index of 0.891 on the test set and 0.903 at the held-out centres. Removing the cardiovascular category changed the concordance by 0.001 (−0.001 to 0.003) and 0.000 (−0.001 to 0.002). Hazard ratios per standard deviation were 1.37 for CDR-SB, 0.76 for the memory score, 1.40 for age and 1.20 for each APOE $\varepsilon 4$ allele, while hypertension (0.995, 0.953–1.038), diabetes and stroke had hazard ratios close to one.

Reversed diagnoses. A clinical diagnosis can be revised. Of the 2,818 converters, 217 (7.7%) were recorded as not demented at a later visit. Excluding them, XGBoost reached an AUROC of 0.947 on the test set and 0.950 at the held-out centres (MCI: 0.839 and 0.851). The conclusions do not depend on reversed diagnoses.

Variables close to the diagnosis. Adding the daily-function questionnaire, independence, memory complaints and Alzheimer's medication increased the AUROC by 0.006 on the test set and 0.005 at the held-out centres. These variables were therefore left out of the main model with little loss of accuracy.

Chapter 7

System Design

The models are packaged as a research prototype for a clinician: an interface for entering a patient's information and seeing a risk estimate with its explanation, and an evidence assistant that answers questions from a fixed library of published papers. The prototype is not intended for clinical use and states this on every screen.

7.1 Architecture

Table 7.1: Components of the prototype.

Component	Role
Inference module (`scripts/inference.py`)	Converts raw entries into the 48 model features (for example a MoCA score into the global z-score), runs the modality-dropout model and the PET fusion model, computes SHAP values and checks the plausibility of the entries
API (`app/api.py`, FastAPI)	Serves the form definition, predictions and evidence-assistant answers
Interface (`app/web`, React)	The form and the results
Evidence assistant (`scripts/rag.py`)	Retrieval, guardrails and answer generation

7.2 Input form

The form is generated from the list of features the model was trained on, so it asks for exactly those inputs and nothing else. It has one section per category: demographics (required), cardiovascular history and measurements, the clinician's assessment (cognitive status and CDR-SB), test scores, genetics, and PET. The clinician marks each optional section as available or not. Empty fields are passed to the model as missing values; no default values are substituted. For each test the form shows the expected score of a healthy person of the same age, sex and education.

7.3 Output

The interface reports the estimated three-year risk next to the observed three-year conversion rate of training participants in the same age group and with the same cognitive status, which gives the clinician an empirical reference point. It lists which categories were used and which were missing, and shows the SHAP contributions of each category and of the most influential individual items. When no cognitive information is available, it states that the estimate is little more than an age-based average.

7.4 Plausibility checks

Contradictory entries can produce estimates that look precise but are meaningless. For example, a MoCA score of 0 entered together with a status of "normal" and a CDR-SB of 0 yields a low risk, because the model gives most weight to the clinician's rating. The interface therefore checks the entries against the ranges observed in training data before showing a result. Entries that contradict each other (test scores below the first percentile of cognitively normal participants combined with a normal status; a CDR-SB above the 99th percentile for the stated status; systolic pressure not above diastolic; implausible smoking duration) are flagged and the estimate is marked as unreliable. Unusual but possible entries (age outside the 1st to 99th percentile of the training cohort, very low education, discordant PET values) generate a note.

7.5 Behaviour checks

Twenty-eight automated checks test the deployed model through the same code path as the interface. They confirm that the form covers every trained feature and no other; that typical cognitively normal, impaired and MCI cases receive plausible, correctly ordered estimates (0.6%, 4.0% and 60.8%); that a worse CDR-SB, a worse cognitive status, lower test scores, more APOE $\varepsilon 4$ alleles or more behavioural symptoms never lower the estimate; that contradictory entries are flagged; and that missing categories are reported. Twenty-seven checks pass. The remaining check compares predicted and observed risk within age and diagnosis groups on held-out data: ten of eleven groups agree within seven percentage points, but for participants with MCI under the age of 65 the model predicts 35% against an observed 27%. This over-prediction is reported as a known limitation.

7.6 Evidence assistant

The evidence assistant is a retrieval-augmented generation system [11]. Its library consists of eleven open-access papers on dementia risk factors, vascular contributions, APOE, amyloid and tau PET and diagnostic criteria, divided into 447 passages. For a question, two rankings are computed, one by semantic similarity of sentence embeddings [12] and one by BM25 keyword relevance [13], and merged by reciprocal rank fusion [14]. If the best semantic similarity is below a fixed threshold (0.35), the assistant reports that the library contains no evidence on the question. Otherwise a small instruction-tuned language model running locally writes an answer from the retrieved passages with numbered citations; if

the answer does not cite the passages correctly, the assistant quotes them directly instead. Questions asking for advice about a particular person are answered with a notice that the system cannot give individual medical advice, followed by general published information only. No data leave the machine.

Table 7.2: Evaluation of the evidence assistant (52 questions).

Measure	Result
Correct source ranked first (hybrid retrieval)	32 of 32
Correct source ranked first (embeddings only / BM25 only)	28 of 32 / 31 of 32
Personal-advice questions flagged	8 of 8
General questions wrongly flagged as advice	0 of 6
Out-of-scope questions declined	6 of 6
Answers written with valid citations by the language model	22% (otherwise quoted)

These results should be read with care. The library is small and the test questions were written by the project team, so retrieval is an easy test. The local language model is small and produced a properly cited answer for only about one question in five; direct quotation is the fallback and is the safer behaviour for medical content.

Chapter 8

Verification and Reproducibility

Clinical prediction studies are vulnerable to subtle errors that inflate performance: the same person in training and test data, information from the future in the features, mislabelled outcomes, or reported numbers that do not match the saved predictions. A separate verification program (`scripts/verify_results.py`) was written to test for these errors independently of the main pipeline. Each check recomputes a quantity in a second, simpler way or tests a property that must hold if the pipeline is correct.

8.1 Checks

Table 8.1: Independent verification checks and their results.

Group	Check	Result
A	Raw files unchanged since the first audit; all 599 image folders present	Pass
B	No participant in more than one partition; no held-out centre in the training data; saved split matches the modelling table	0 participants and 0 centres shared
C	Features at the first and third visit are identical when all later visits are deleted before the features are built	Identical for 1,500 participants × 67 features at each visit
C	No PET value is newer than the visit at which it is used	0 of 1,481 amyloid and 713 tau values
D	Outcome labels recomputed with an independent implementation	0 disagreements in 11,403 visits
D	No eligible index visit has a dementia diagnosis	0 of 134,943
E	With shuffled outcome labels, a retrained model falls to chance	AUROC 0.508
F	Reported AUROC values recomputed from the saved predictions	Largest difference 0 over 5 models and 2 evaluation sets
F	Cohort sizes in the reports match the modelling table	Match
G	Retraining from scratch with a new seed reproduces the result	Test 0.937, held-out 0.946 (reported 0.938, 0.946)
G	Gain from the cardiovascular category with a fresh seed	+0.0016
H	Cohort flow and PET index counts in the reports match the data	Match

All seventeen checks passed on the final pipeline (re-run on 22 October 2026).

8.2 Errors found and corrected

The checks found two genuine errors in an earlier version of the pipeline, both of which used information that would not have been available at the time of prediction:

1. Missing education and race at an early visit had been filled in from a later visit for about 0.1% of participants. Fixed characteristics are now carried forward only.

2. PET scans had been matched to the nearest visit, which for many participants was a visit a few weeks before the scan. A scan is now used only at a visit on or after its acquisition date.

After both corrections every model was retrained and every result regenerated. The main results changed only in the third decimal place. Earlier outputs were retained for comparison.

8.3 Reproducibility

All random seeds are fixed (2026 for the split, 0 to 2 for model training). Package versions are recorded in `requirements.txt`. The pipeline is run by a fixed sequence of scripts documented in the repository; given the NACC data, it regenerates every table in this thesis. Participant-level data and trained models are not distributed, in accordance with the data use agreement; all shared outputs are aggregates, and each report is checked automatically for participant identifiers before it is written.

Chapter 9

Discussion and Limitations

To be written (see docs/project_ audit.md).

Chapter 10

Conclusion and Future Work

To be written.

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